Final Course Project

AI-Assisted Release Check & Ownership Evidence

Project scenario

You are preparing your Task Tracker for a teammate who will maintain it after the course. The teammate should be able to clone the repo, run the app locally, run the tests, run the app in Docker, understand the main release checks, and see how AI was used responsibly.

Your final project should answer four questions:

- Does the existing Task Tracker still work and stay inside the intended scope?
- Can another developer run the app, tests, CI workflow, and Docker container from the repo instructions?
- Can you show evidence that AI output was reviewed, graded, corrected, or rejected instead of accepted blindly?
- Can you state the rules you will use when working with AI after the course?

Starting point and ground rules

Rule	What it means
Use the same repo	Use the public GitHub repository used for the mid-course project and later modules.
Work on final-project	All final work should live on one branch named final-project.
No new product features	Do not implement comments, authentication, production database, notifications, or unrelated UI changes.
Protect app/ and frontend/	Only change app/ or frontend/ for a small bug fix, security fix, or documentation-supported correction. Explain any such change in docs/final-ai-review.md.
No real secrets or personal data	Do not paste credentials, .env values, tokens, production logs, or real personal/customer data into AI tools or the repo.
You own the result	If you cannot explain a changed line, command, config choice, or AI suggestion, do not submit it as final work.

Deliverables at a glance

Area	Required evidence	What graders check
Baseline and scope control	README Final Project section; release-evidence baseline; final-project branch.	The app still runs, tests pass or are honestly recorded, and no new product feature was added.
Release readiness	.github/workflows/ci.yml, Dockerfile, .dockerignore, docs/release-evidence.md.	CI runs pytest, Docker builds/runs, /health is verified, and dangerous shortcuts are not used.
AI review and security	AGENTS.md and docs/final-ai-review.md.	AI comments/findings are graded with reasons, file

Area	Required evidence	What graders check
		evidence, and at least one human/manual check.
AI ownership	docs/ai-playbook.md plus ownership statement in docs/final-ai-review.md.	Rules are concrete, personal, and backed by actual course evidence.
Submission	Public GitHub URL only.	No video link, no zip file, and no separate screenshots are required unless the LMS asks for them.

Required final repository structure

Your repo may contain more files, especially if you completed all Module 4 and Module 5 exercises. For the final project grade, the minimum required structure is:

```
.github/workflows/ci.yml
Dockerfile
.dockerignore
README.md
AGENTS.md
app/
frontend/
tests/
docs/
  release-evidence.md
  final-ai-review.md
  ai-playbook.md
```

Part A - Baseline app check and README

Start by proving the current app still works. Do not ask AI to rewrite the project. The first responsibility of the final project is to protect what already exists.

Requirement	What to submit
Final branch	All final project work is on a branch named final-project.
Backend baseline	Record the command used to start the API and the result of GET /health.
Frontend baseline	State how you opened the frontend and one sentence confirming the Kanban board/create-edit flow is still visible. No video is required.
Test baseline	Run the full pytest suite and record the exact command and result. If tests fail, record the failing test names and whether the failure is pre-existing or introduced by final work.
README Final Project section	Add branch name, local run command, Docker run command, test command, evidence files, and a short AI-assistance summary.

Part B - Release readiness evidence

Save all evidence for this part in docs/release-evidence.md. Keep it factual and short. Text logs are enough; screenshots are optional.

B1. Continuous Integration

- Create or finalize .github/workflows/ci.yml so pytest runs on push and pull request.
- Record the latest green GitHub Actions run link or a short note that the workflow ran successfully.

- Check for dangerous shortcuts: continue-on-error, || true, skipped pytest command, vague Python version, or missing dependency installation.
- Intentional red-run evidence is optional for the final project. Include it only if you already produced it during Module 4.

B2. Docker and runtime verification

- Create or finalize Dockerfile and .dockerignore.
- Build and run the image locally.
- Verify the container responds to /health with HTTP 200.
- Record a short Docker safety check: non-root user if implemented, no .env/secrets copied, and a clear runtime command.

B3. Documentation checked against reality

- Update README.md with exact setup, run, test, and Docker commands.
- Check at least three README or AI-generated documentation claims against the actual repo or running app.
- At least one checked claim should involve a command, endpoint, schema/status code, Docker behavior, or CI behavior.

Part C - Final AI review and security evidence

Save this part in docs/final-ai-review.md. This is the main AI Assisted Coding evidence document. It should be short enough to grade quickly but specific enough to show judgment.

Section	Minimum requirement
AGENTS.md guardrails	Confirm AGENTS.md exists and includes stack, run/test commands, project rules, and docs-first/read-first guardrails.
AI code review mini-log	Choose one real diff or changed file. Record at least 3 AI review comments and grade each Useful, Noise, or Wrong with a reason.
AI security mini-review	Run or reuse a read-only security review. Record at least 3 findings with file evidence and grade each Valid, False Positive, or Noise.
Manual check	Add one human check that did not simply copy the AI output. This may be a manual finding or a note that you checked a specific risk and found nothing new.
Rejected or corrected AI output	Describe one AI suggestion you rejected, corrected, downgraded, or refused to apply.
Ownership statement	Write 3-5 sentences explaining why you are comfortable submitting this repo as your own work.

What counts as good evidence?

Good evidence names real files, real commands, real test results, real grades, and real decisions. Weak evidence says only “AI helped” or “I checked it” without saying what was checked.

Part D - Personal AI playbook

Save or update docs/ai-playbook.md. This may be the same playbook you created in Module 5, revised after finishing the final project.

Required section	What graders look for
When I reach for AI first	Specific task shapes where AI helped you during the course.
When I do not reach for AI first	Situations where risk, missing context, or learning goals should slow you down.
My non-negotiables	Rules around secrets, data, review, verification, and ownership.
My review rules	How you inspect diffs, run commands, grade findings, and reject bad suggestions.
What I am still figuring out	Honest open questions about tool choice or team norms.
Decision Card	New feature, code review, debugging, infrastructure, never-paste, and one rule.

Keep the playbook to one page. It should sound like you, not like a vendor policy or generic AI safety statement.

Recommended workflow for students

1. Create and push the final-project branch.
2. Run the app and tests before final edits. Record the baseline in docs/release-evidence.md.
3. Finish or clean up CI, Docker, and README.
4. Complete docs/release-evidence.md using the template below.
5. Create or refine AGENTS.md with repo-specific guardrails.
6. Run a small AI code review and security review, then complete docs/final-ai-review.md.
7. Revise docs/ai-playbook.md after the final evidence exists.
8. Check the quick grading checklist, then submit the GitHub URL.

README Final Project section template

```
## Final Project

Branch reviewed: final-project

### What this submission demonstrates
- Existing Task Tracker app still runs inside the intended course scope.
- CI runs the pytest suite on push and/or pull request.
- Docker image builds and runs with /health returning 200.
- AI review, security, and ownership evidence is in docs/.

### How to run locally
[paste exact setup and run commands]

### How to run tests
[paste exact test command]

### How to run with Docker
[paste exact build, run, and curl commands]

### Evidence files
- docs/release-evidence.md
- docs/final-ai-review.md
- docs/ai-playbook.md

### AI assistance summary
AI helped draft or review: [CI / Docker / docs / security / debugging].
I verified the work by: [tests / diff review / Docker / /health / manual scan].
One AI suggestion I rejected or corrected: [brief note].
```

docs/release-evidence.md template

```
# Release Evidence

## Baseline
- Branch:
- Date:
- Local app run command:
- /health result:
- Frontend check:
- Test command:
- Test result:

## CI evidence
- Workflow file:
```

```
- Latest run link or note:
- Test command used by CI:
- Shortcut check: no continue-on-error / no || true / pytest is not skipped.

## Docker evidence
- Build command:
- Run command:
- /health check:
- Non-root check, if implemented:
- No-baked-secrets check:

## Documentation claim-vs-reality log
| Claim checked | Evidence used | Result | Change made, if any |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
```

docs/final-ai-review.md template

```
# Final AI Review and Ownership Evidence

## AGENTS.md guardrails
- Repo-specific stack and commands included: yes/no
- Docs-first/read-first guardrail included: yes/no
- Unexpected app/frontend edits rule included: yes/no

## AI code review mini-log
| AI comment | Grade: Useful / Noise / Wrong | Reason | Verification or decision |
|---|---|---|---|
| | | | |
| | | | |
| | | | |

## AI security mini-review
| Finding | File evidence | Grade: Valid / False Positive / Noise | Reason | Next action |
|---|---|---|---|---|
| | | | | |
| | | | | |
| | | | | |

## Manual security check
[What I checked myself, what I found, and why it matters.]

## One AI output I rejected or corrected
[What AI suggested, why I did not accept it as-is, and what I did instead.]

## Three AI usage rules
1. Never paste:
2. Always verify:
3. Record AI contributions by:

## Ownership statement
[3-5 sentences explaining why I am comfortable submitting this repo as my work.]
```